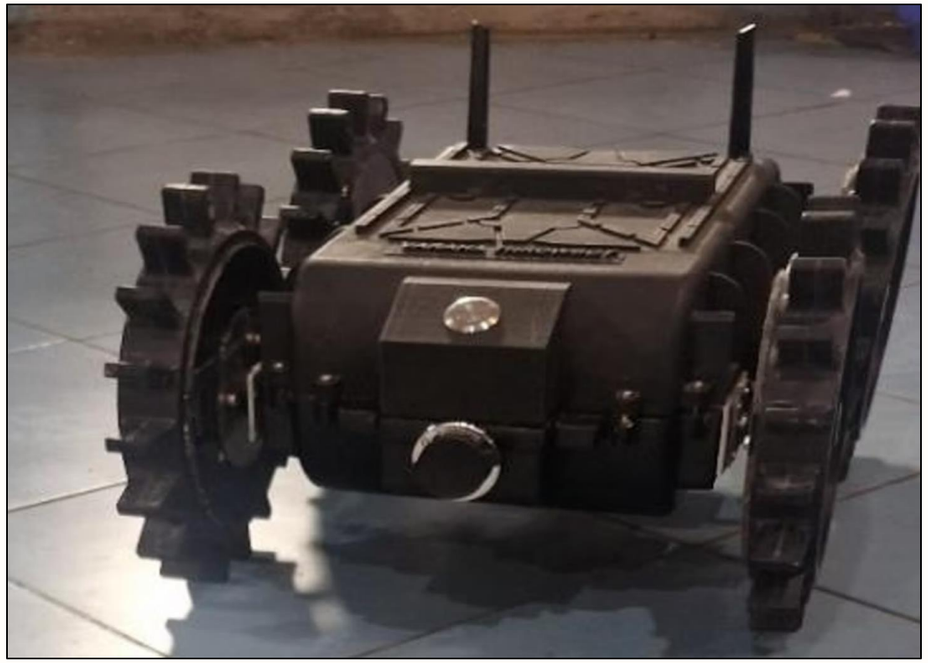
Unveiling the DNS: Autonomous Surveillance

-By Janyu Tech

Overview

This presentation introduces the Day-Night Surveillance Bot (DNS) by Janyu Technologies - a compact, versatile Unmanned Ground Vehicle (UGV) built for all terrain maneuvering, real-time monitoring and support in defense and security operations. Designed to operate in challenging environments, DNS plays a critical role in remote surveillance, area inspection, and mobility-driven patrolling.

With a modular design and adaptable structure, DNS is engineered for mission flexibility, enabling both manual and semi-autonomous deployment in a wide range of scenarios.



Purpose

Rugged Mobility

DNS is a 2-wheel drive UGV designed for remote, rugged, day-night surveillance across diverse terrains.

Strategic Applications

Ideal for border patrol, security monitoring, and industrial hazard zone inspection.

Advanced Imaging

Equipped with real-time HD camera, IR night vision, and GUI-based navigation for comprehensive visibility.

Personnel Safety

Enables reconnaissance and operation in hazardous areas without exposing human personnel to threats.

JANYU TECH

DNS – Core Specifications

- **Dimensions:** 230 mm (W) × 180 mm (H)
- **Weight:** 5 kg
- **Drive System:** 2WD – 210 mm rubber wheels
- **Terrain Capability:** 30° slopes, 150 mm stairs
- **Max Speed:** 3–4 km/h (adjustable)
- **Battery Runtime:** 3–4 hours
- **Charging Time:** 2.5 hours
- **Protection:** IP67 (dust and water immersion), MIL-STD-810 (environmental resilience)

Key Features

- All terrain mobility.
- Vision and Imaging system.
- Wifi connectivity.
- Remote monitoring and control.
- Real-time video streaming.
- High endurance Battery backup.

JANYU TECH

Manufacturing and Materials

1 Chassis Durability

Constructed from durable ABS material a compact body for maximum resilience and protection in harsh environments.

2 All-Terrain Wheels

Equipped with 210 mm rubberized outdoor treads designed for superior grip and maneuverability across diverse terrains.

3 Precision Mechanics

Utilizes robust rubber belts, and idler wheels for smooth, reliable power transmission.

4 Vibration Control

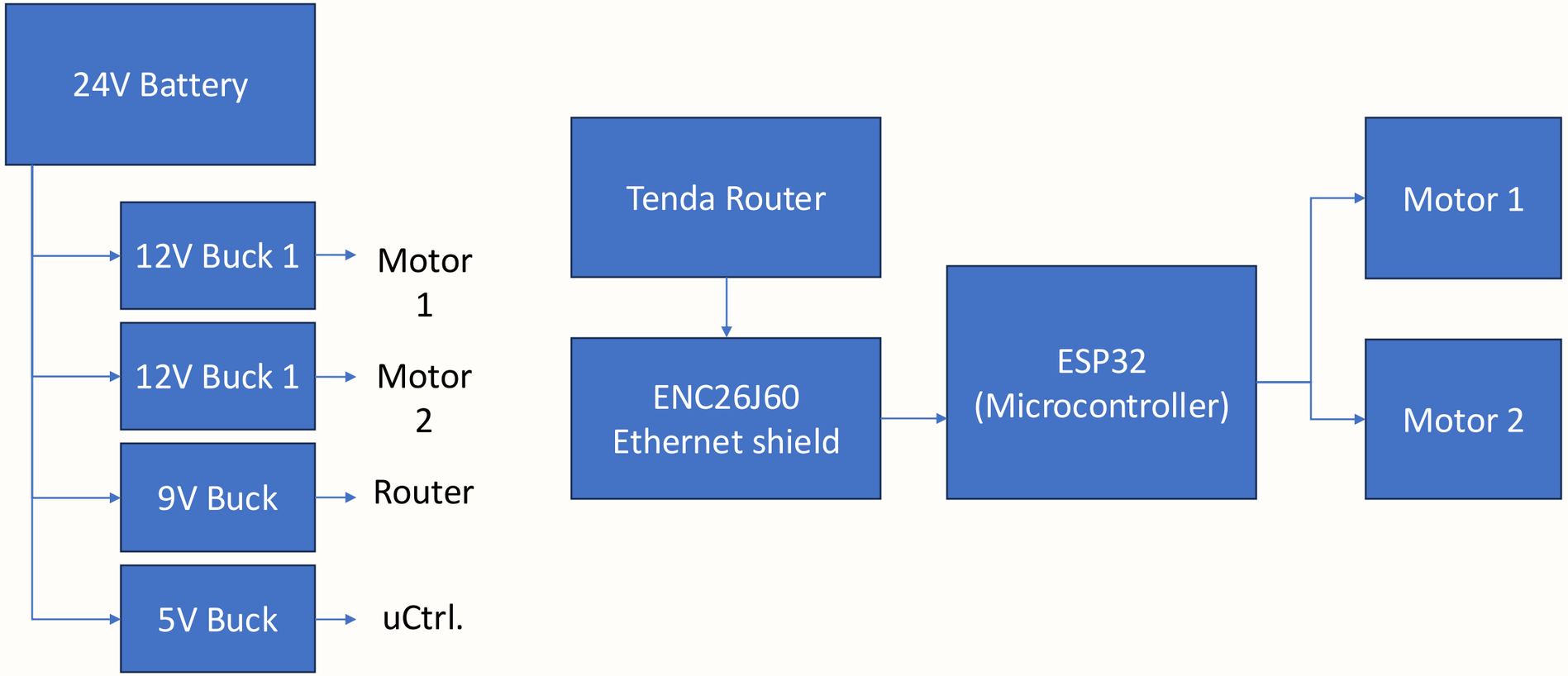
Electronics are mounted on an isolated, vibration-dampened layout, ensuring stable performance and longevity.

5 Modular Design

Features easy access points for power, motor, and logic sections, facilitating maintenance and future upgrades.

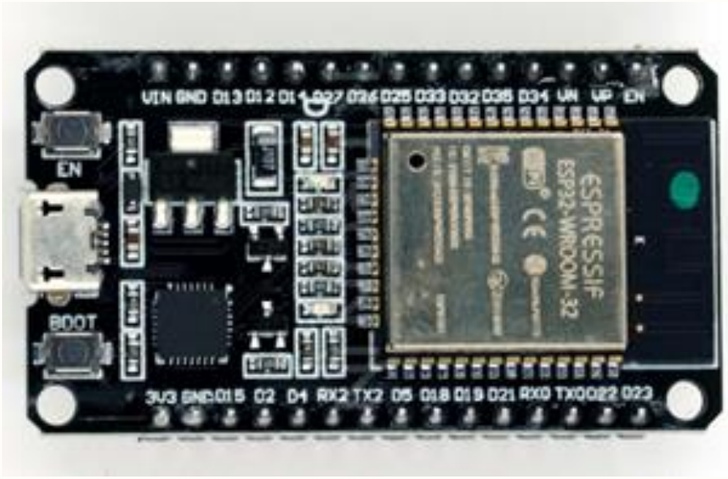
JANYU TECH

Block diagram



Components

- **The Brain**



ESP32 Board

Parameter	Specification	Details
Microcontroller	ESP32 (ESP-WROOM-32)	Dual-core Xtensa LX6 (32-bit)
Clock Speed	Up to 240 MHz	Adjustable
Wireless	Wi-Fi 802.11 b/g/n (2.4 GHz), Bluetooth 4.2	Supports BLE/Mesh
Operating Voltage	3.3V	Input: 5V via USB
I/O Pins	30+ digital, 15+ analog (12-bit ADC)	2x 8-bit DAC
Memory	4MB Flash, 520KB SRAM, 448KB ROM	8KB RTC Fast Memory
Power Consumption	80–100 mA (active), 10 µA (deep sleep)	Wi-Fi/BT enabled
Dimensions	55mm × 28mm	Weight: 10–15g
Applications	IoT, robotics, wearables	Audio/voice projects

JANYU TECH

Components

- **Ethernet Shield**

- ENC28J60 Module

Parameter	Specification	Details
Controller	ENC28J60 (Microchip)	SPI interface
Speed	10 Mbps (IEEE 802.3)	Half/Full duplex
Voltage	3.3V	200 mA active current
Interface	SCK, SO, SI, CS, INT, RST	RJ45 with magnetics
Protocols	TCP/IP, UDP, DHCP	Hardware CRC/DMA
Applications	IoT gateways, embedded servers	Industrial control

JANYU TECH

Components

- **Motors**

- Faulhaber Motors with Gear box

Parameter	Motor	Gearhead	Combined System
Nominal Voltage	12V	—	—
No-Load Speed	6,300 rpm	—	**95 rpm** (43:1 ratio)
Rated Torque	33.5 mNm	1.5 Nm	**1.224 Nm**
Peak Torque	67 mNm (≤5s)	4.5 Nm	**2.45 Nm** (short-term)
Efficiency	95% (electronic)	85%	**52.1%** (system)
Dimensions	32mm diameter, 61.5mm length	26mm diameter	Compact integration
Applications	Precision robotics	High-torque actuators	CNC, medical devices

JANYU TECH

Components

- **Battery – Li-ion**

Parameter	Specification	Details
Chemistry	Lithium-Ion (7S)	BMS included
Energy	242 Wh	25.2V × 9.6Ah
Dimensions	85mm × 58mm × 127mm	Weight: ~1.5kg
Cycle Life	500–1,000 cycles	80% capacity retention
Charging Voltage	29.4V (CC/CV)	4.2V per cell
Discharge Rate	10A continuous	15–20A peak

JANYU TECH

Components

- SMPS Charger

- SMPS Charger (29.4V, 4A)

Parameter	Specification	Details
Input	180–280V AC	Global compatibility
Output	29.4V DC, 4A	117.6W power
Charging Time	~2.5 hours	9.6Ah / 4A
Protections	OVP, short-circuit, thermal	Passive/active cooling
Compliance	CE, RoHS, IEC 62368	Safety-certified

JANYU TECH

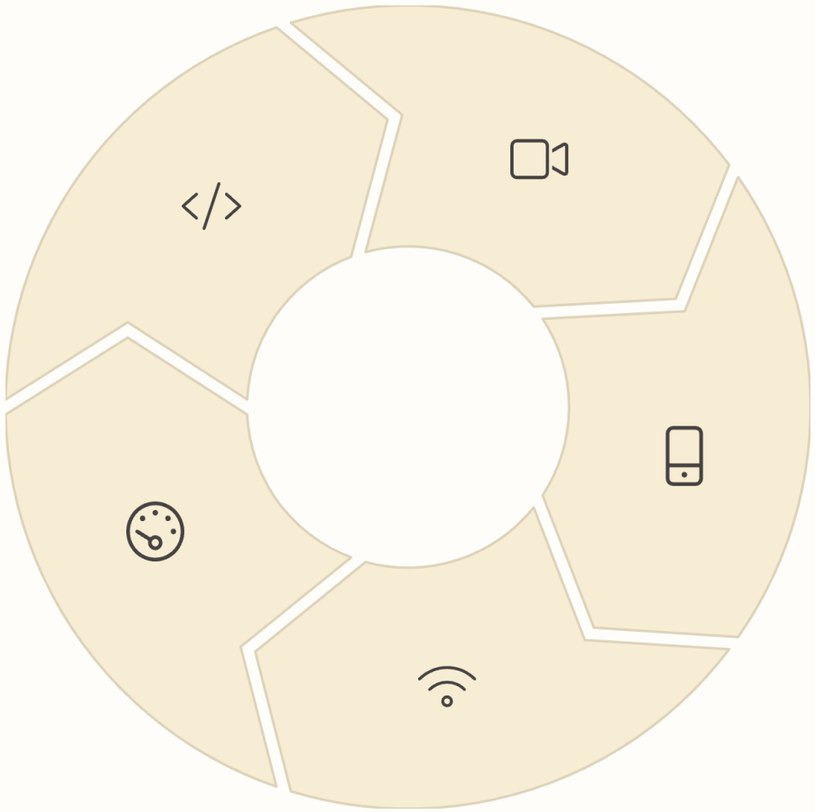
Software and Coding – The Brains

C Firmware

Robust control logic on ESP32 for real-time motor and system management.

PWM Motor Control

Precise, real-time adjustments based on GUI commands for optimal maneuverability.



RTSP Camera Stream

High-definition video feed accessible via Android GUI and desktop applications.

Android GUI

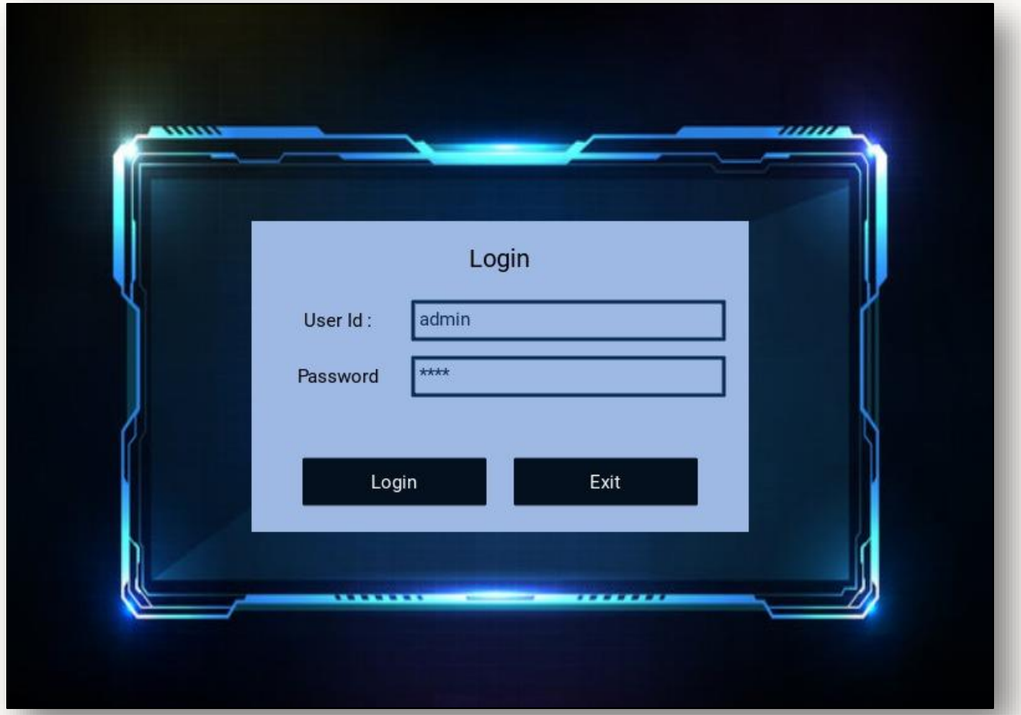
Intuitive mobile app for navigation, speed control, and operational logs.

Dual Communication

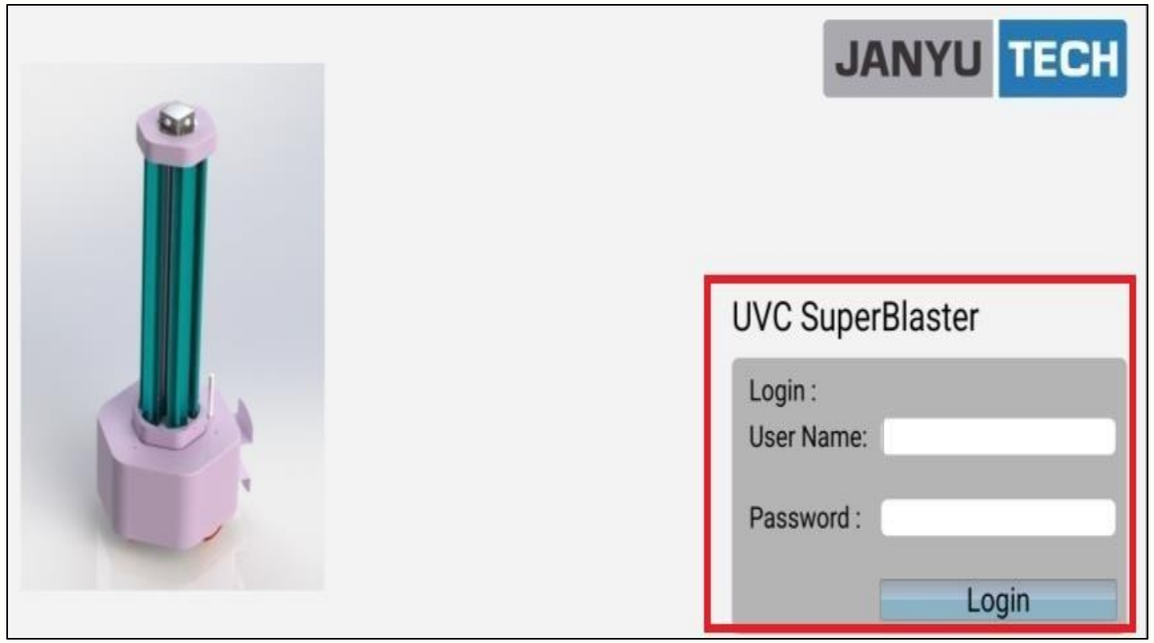
Primary Wi-Fi and backup Ethernet ensures reliable connectivity.

JANYU TECH

Graphical User Interface (GUI Features)



Laptop App



Android App

Graphical User Interface (GUI Features)

1

Secure Login

Access control requires credentials before initiating robot operation.

2

Intuitive Drive Page

Direct access to essential controls: Robo Logs, Settings, and Exit functions.

3

Dynamic Wi-Fi Status

'Searching robot' status transitions to 'Connected' upon successful signal detection.

Graphical User Interface (GUI Features)

4

Adjustable Speed Control

Select between Slow, Medium, and Free Speed modes, with Slow as the default setting.

5

Precision Navigation

A dedicated widget enables seamless Left-Right and Forward-Backward crawler movement with live camera feed using RTSP protocol.

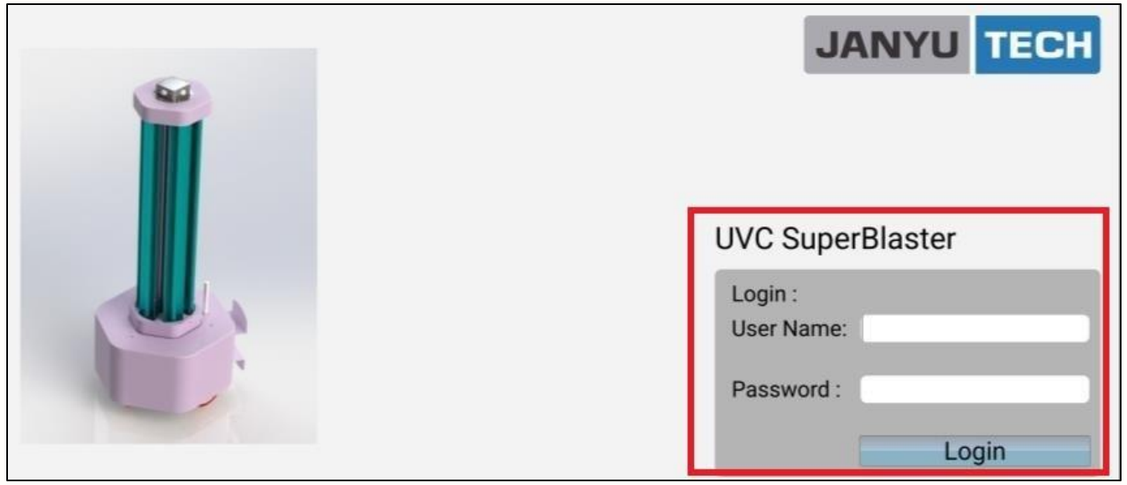
6

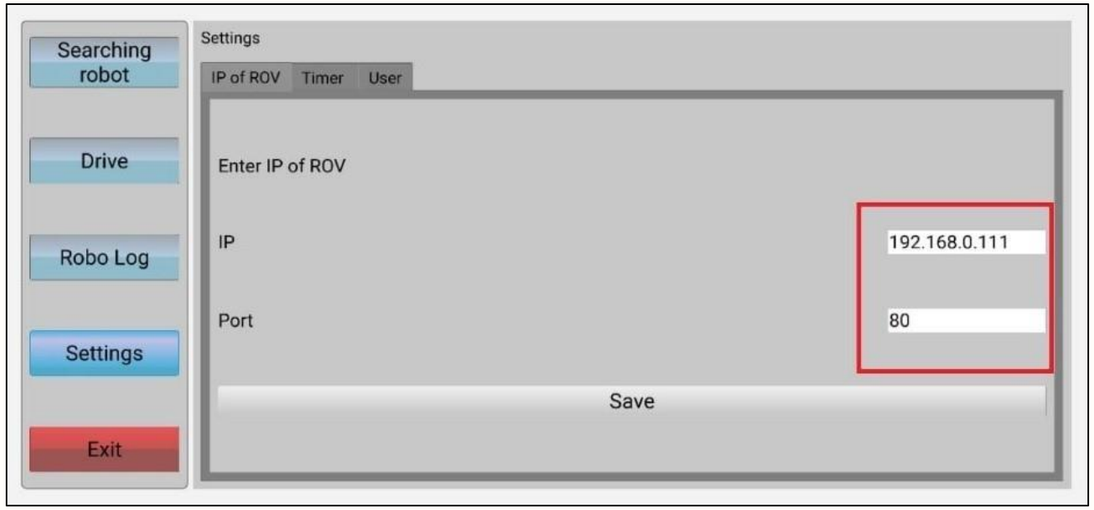
Comprehensive System Logs

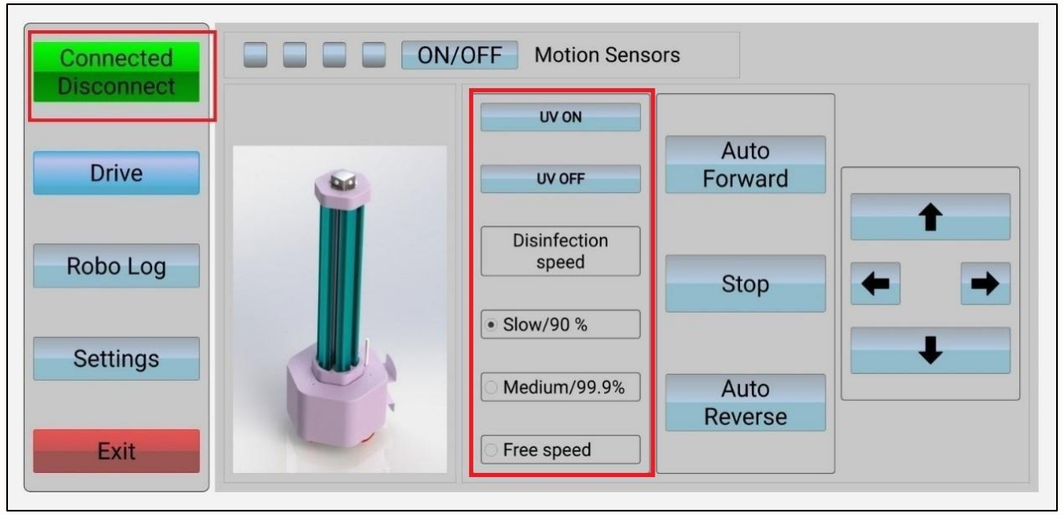
Detailed operational history is recorded and accessible in the Robo Logs section for review.

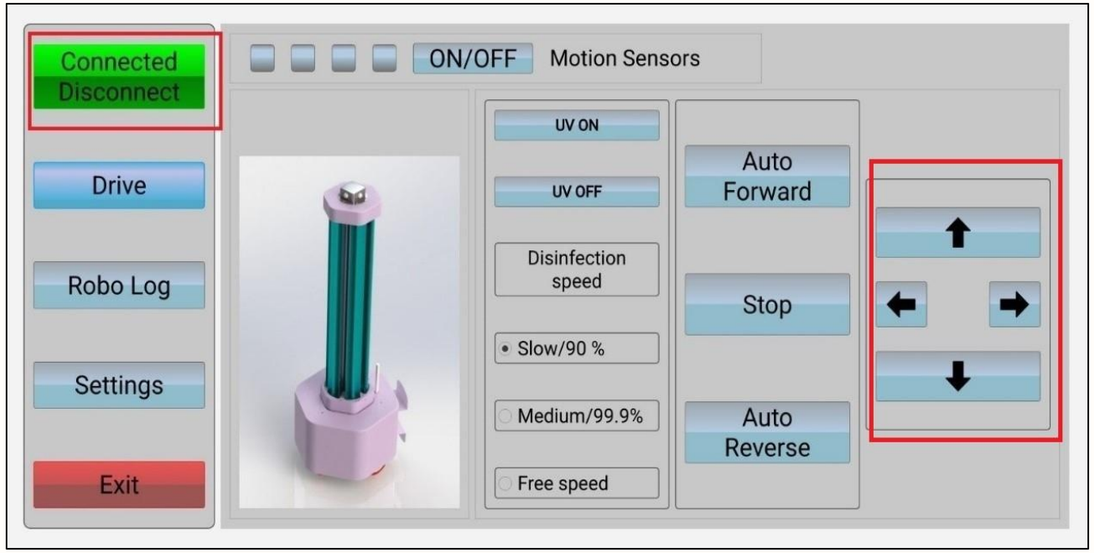
JANYU TECH

Graphical User Interface (ANDROID)









JANYU TECH

Camera Access on Desktop

1

RTSP Stream Integration:

The system retrieves the camera feed from a real-time RTSP link configured within the GUI

2

Integrated Feed Display:

Live visuals appear at the center of the GUI, allowing the user to stay focused on one control screen.

3

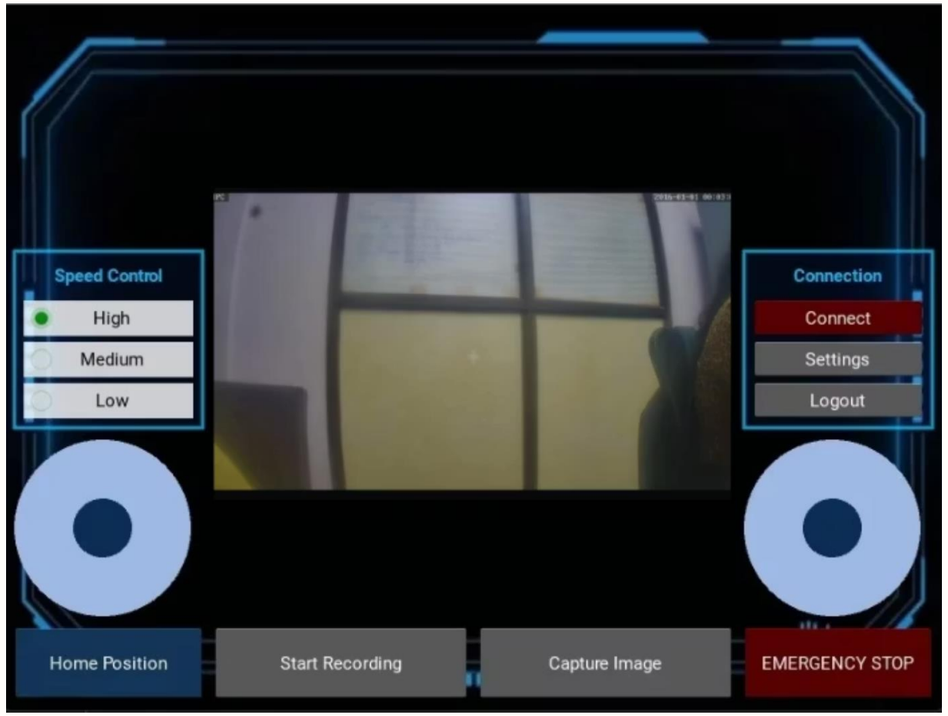
Live Situational Awareness:

Operators monitor terrain, movement paths, or room activity for obstacle detection and safe routing. Operators monitor terrain, movement paths, or room activity for obstacle detection and safe routing.

4

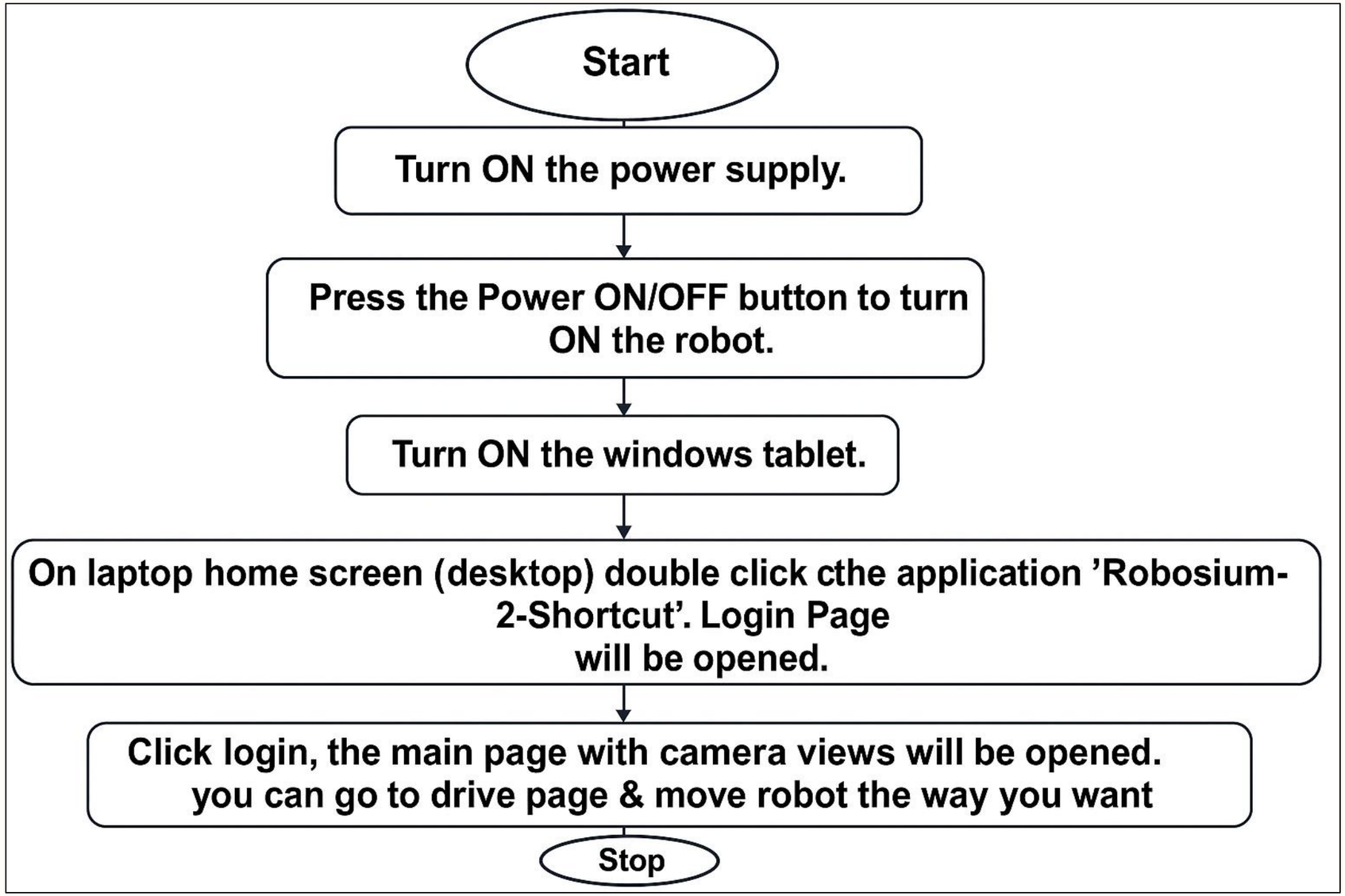
Unified Control & Vision:

Everything—video, controls, speed, and navigation—is managed from one GUI



JANYU TECH

Operational Flowchart



JANYU TECH

Testing & Field Trials

The DNS has undergone extensive real-world trials to validate its ruggedness, usability, and field readiness, ensuring peak performance in diverse environments.

Terrain Testing

- **Smooth Floor Navigation:** Seamless roving on ceramic tiles, wooden floors, and epoxy-coated labs, demonstrating precise indoor maneuverability.
- **Rough Indoor Terrain:** Operational across uneven surfaces, metal ramps, and debris-like patches in test facilities, proving adaptability to challenging ground.
- **Incline Test:** Successfully climbed slopes up to 30° and navigated stairs up to 150 mm height, validating its robust climbing capabilities.

JANYU TECH

Testing & Field Trials

The DNS has undergone extensive real-world trials to validate its ruggedness, usability, and field readiness, ensuring peak performance in diverse environments.

Functional Testing

- **Live Video Feed Stability:** Maintained no feed drops during 3-hour continuous roving under Wi-Fi control, ensuring uninterrupted surveillance.
- **GUI Responsiveness:** Provided instant feedback for speed control, operational logs, and joystick inputs under various test latencies, confirming user-friendly control.

- **Battery Endurance:** Sustained 3.5 hours of runtime under mixed-load navigation with full IR vision, surpassing operational requirements.

JANYU TECH

Safety Systems & Fail-Safes

The DNS is designed with robust safety protocols to ensure reliable operation in harsh and unpredictable environments, protecting both personnel and assets.

Emergency Stop Button

Instantly cuts power to motors and electronics in case of malfunction or human error, providing immediate shutdown.

System Status Indicators

LEDs provide clear visual status: green for normal, red for fault, and blinking for standby or signal issues.

Overvoltage & Overcurrent Protection

Integrated buck converters maintain voltage within safe operating limits, preventing damage to sensitive components.

Signal Loss Handling

In the event of Wi-Fi disconnection, DNS automatically switches to idle mode to prevent uncontrolled or unintended movement.

Operator Override via GUI

Manual intervention is always possible through the GUI with stop/halt commands or speed throttling capabilities.

JANYU TECH

Troubleshooting & Limitations

Understanding known constraints ensures smoother operation and safe handling of the DNS bot.

Known Limitations

Wi-Fi Dependence

Operates best within 30–40m line-of-sight; walls or interference can significantly reduce range.

No Obstacle Avoidance

Operator must manually guide the bot around obstacles; autonomous avoidance requires future sensor integration.

RC-Based Control

Entirely remote control-based; advanced path planning or object recognition needs future ML integration.

Not Waterproof

Unsuitable for wet or slippery surfaces; avoid moisture-heavy environments to prevent damage.

Quick Fixes

1

RTSP Feed Issues

Recheck IP and port settings; ensure the camera has power and is properly connected.

2

Bot Not Moving

Confirm the GUI is connected, check battery levels, and verify the emergency stop button is disengaged.

3

Control Lag

Reduce network congestion or bring the operator device closer to the bot to improve signal.

JANYU TECH

Deployment Scenarios

The DNS robot's robust design and versatile capabilities enable rapid deployment across various critical real-world environments without custom setups.

Military Training

Provides mobile surveillance within training compounds for enhanced situational awareness.

Industrial Zones

Monitors equipment in hazardous or restricted areas like power plants and oil fields.

Disaster Response

Conducts initial assessments in areas affected by earthquakes or structural collapses.

Warehousing

Enables live night monitoring of automated robotic aisle zones for security.

Railway Security

Offers critical surveillance in tunnels, under platforms, and luggage areas.

JANYU TECH

Use Cases of DNS

DNS's adaptability makes it suitable for diverse field operations, providing critical support in varied environments.

Border Surveillance

Patrolling sensitive outdoor areas with night vision and remote feed capabilities.

Industrial Monitoring

Inspecting remote or hazardous factory zones without risking human personnel.

Emergency Response

Conducting reconnaissance in collapsed buildings or after natural disasters.

Perimeter Security

Providing live surveillance around military bases or testing grounds.

Nighttime Patrolling

Enabling IR-enhanced inspection during blackouts or low-light operations.

Quarantine Zone Support

Offering visual monitoring in biohazard or restricted access zones.

JANYU TECH

Future Upgrades & Roadmap

Thermal Camera

Heat signature detection in low-visibility.

LiDAR 3D Mapping

Real-time spatial awareness and navigation.

AI Navigation

Autonomous route planning and obstacle avoidance.

Cloud & IoT

Remote data storage and bot control.

Swarm Coordination

Multi-bot teaming for large area coverage.

JANYU TECH

Conclusion: Empowering Tomorrow's Defense

This Day-Night Surveillance Bot (DNS) by Janyu Tech represents a significant leap in autonomous defense. Designed for extreme reliability and operational flexibility, DNS empowers security forces with unparalleled situational awareness in the most challenging environments.

Unrivaled Reliability

Robust design and MIL-STD-810 compliance ensure peak performance in harsh conditions.

Superior Vision

HD night vision and RTSP integration provide comprehensive visibility, day or night.

Intelligent Operations

Intuitive GUI and advanced firmware enable precise control and tactical advantage.

Future-Ready

Roadmap includes AI navigation, LiDAR mapping, and swarm coordination for evolving needs.

Janyu Technologies is committed to continuous innovation, ensuring DNS remains at the forefront of defense surveillance technology.

JANYU TECH

Frequently Asked Questions

JANYU TECH